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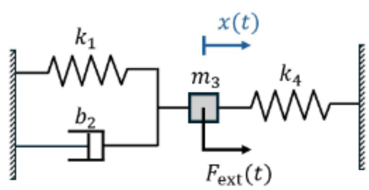
Aero 4630

Professor Luo

Homework 4

Consider the following diagram with mass, damper, and springs. The constitutive law constants are k_i, m_i, and b_i, where i stands for the index of the component. All the following problems are based on this system.



1. Modeling systems with 2nd order ODE
   a. Write out the newtons laws without plugging in constitutive laws of this system.

$$-f_{k1} - f_{b2} + k_4 + f_{ext} = m_3 a_3$$

   b. Write out all the kinematic constraint equations.

$$v_1 = v_2 = v_3 - 0$$

$$v_4 = 0 - v_3$$

$$v_1 = v_2 = v_3 = -v_4$$

   c. Write out the constitutive laws for each component

$$f_{k1} = k_1 x_1; f_{b2} = b_2 \dot{x}_2; f_{k4} = k_4 x_4$$

   d. Derive the second order ODE for this system

$$-k_1 x_1 - b_2 \dot{x}_2 + k_4 x_4 + f_{ext} = m_3 \ddot{x}_3$$

$$-k_1 x_3 - b_2 \dot{x}_3 - k_4 x_3 + f_{ext} = m_3 \ddot{x}_3$$

$$\ddot{x}_3 = -\frac{k_1}{m_3} - \frac{b_2}{m_3}\dot{x}_3 - \frac{k_4}{m_3}x_3 + \frac{f_{ext}}{m_3}$$

$$\ddot{x}_3 = -\frac{b_2}{m_3}\dot{x}_3 - \frac{k_4 + k_1}{m_3}x_3 + \frac{f_{ext}}{m_3}$$

2. When $f_{ext} = 0, m_3 = 1\ and\ k_1 = k_4 = 1$, find the solution to the governing ODE for the following cases:

$$\ddot{x_3} = -\frac{b_2}{1}\dot{x}_3 - \frac{2}{1}x_3 + 0$$

   a. $b_2 = 2, x(0) = 1\ and\ \dot{x}(0) = 0$; combine multiple trigonometric functions in your result into a single one.

$$let\ x = Ae^{\lambda t}; \dot{x} = \lambda Ae^{\lambda t};\ \ddot{x} = \lambda^2 Ae^{\lambda t}$$

$$\lambda^2 Ae^{\lambda t} = -\frac{2}{1}\lambda Ae^{\lambda t} - \frac{2}{1}Ae^{\lambda t}$$

$$\lambda^2 = -\frac{2}{1}\lambda - \frac{2}{1}; \lambda^2 + 2\lambda + 2$$

$$\lambda = \frac{-2 \pm \sqrt{4 - 4 \cdot 2}}{2} = -1 \pm i$$

$$x = A_1 e^{-t} e^{it} + A_2 e^{-t} e^{-it}$$

$$x = e^{-t}\left(A_1 e^{it} + A_2 e^{-it}\right)$$

$$x = e^{-t}\left(A_1 \cos(t) + A_1 isin(t) + A_2 \cos(-t) + A_2 isin(-t)\right)$$

$$x = e^{-t}\left(A_1 \cos(t) + A_1 isin(t) + A_2 \cos(t) - A_2 isin(t)\right)$$

$$x = e^{-t}\left((A_1 + A_2)\cos(t) + (A_1 i + A_2 i)\sin(t)\right)$$

$$x = e^{-t}\left(Ccos(t) + Dsin(t)\right)$$

$$x(0) = e^0(C + 0) = C = 1$$

$$\dot{x} = (-Ce^{-t}\cos(t) - Ce^{-t}\sin(t)) + (-De^{-t}\sin(t) + De^{-t}\cos(t))$$

$$\dot{x}(0) = (-1 - 0) + (0 + D) = 0; D = 1$$

$$x(t) = x = e^{-t}\left(cos(t) + sin(t)\right)$$

b. $b = 2\sqrt{2}, x(0) = 0\ and\ \dot{x}(0) = 1$

$$let\ x = Ae^{\lambda t}; \dot{x} = \lambda Ae^{\lambda t};\ \ddot{x} = \lambda^2 Ae^{\lambda t}$$

$$\ddot{x_3} + \frac{2\sqrt{2}}{1}\dot{x}_3 + \frac{2}{1}x_3 = 0$$

$$\lambda^2 + 2\sqrt{2}\lambda + 2 = 0$$

$$\lambda = \frac{-2\sqrt{2} \pm \sqrt{8-8}}{2} = -\sqrt{2} \pm 0$$

Repeated roots general soln: $Ae^{\lambda t} + Bte^{\lambda t}$

$$x = \ Ae^{-\sqrt{2}t} + Bte^{-\sqrt{2}t}$$

$$x(0) = A = 0$$

$$\dot{x} = Be^{-\sqrt{2}t} - Bte^{-\sqrt{2}t}$$

$$\dot{x}(0) = B = 1$$

$$x = te^{-\sqrt{2}t}$$

c. $b_2 = 3, x(0) = 2\ and\ \dot{x}(0) = \ -3$

$$\ddot{x_3} + \frac{3}{1}\dot{x}_3 + \frac{2}{1}x_3 = 0$$

$$\lambda = \frac{-3 \pm \sqrt{9-8}}{2} = -\sqrt{2} \pm \frac{1}{2}$$

Overdamped general solution: $Ae^{\lambda_1 t} + Be^{\lambda_2 t}$

$$x = Ae^{(-\sqrt{2}+1/2)t} + Be^{(-\sqrt{2}-1/2)t}$$

$$\dot{x} = A\left(-\sqrt{2} + 1/2\right)e^{(-\sqrt{2}+1/2)t} + B\left(-\sqrt{2} - 1/2\right)e^{(-\sqrt{2}-1/2)t}$$

$$x(0) = 0; A + B = 0; A = -B$$

$$\dot{x}(0) = A\left(-\sqrt{2} + 1/2\right) + B\left(-\sqrt{2} - 1/2\right) = 1$$

$$\dot{x}(0) = A\left(-\sqrt{2} + 1/2\right) + A\left(\sqrt{2} + 1/2\right) = 1; A = 1; B = -1$$

$$x = e^{(-\sqrt{2}+1/2)t} - e^{(-\sqrt{2}-1/2)t}$$

3. Forced oscillation
   a. When $F_{ext} = \sin(3t)\,, m = 1, b_2 = 1\ and\ k_1 = k_4 = 1$, find the general solution to the governing ODE.

$$\ddot{x_3} = -\frac{b_2}{m_3}\dot{x}_3 - \frac{k_4 + k_1}{m_3}x_3 + \frac{f_{ext}}{m_3}$$

$$\ddot{x_3} + \frac{1}{1}\dot{x}_3 + \frac{2}{1}x_3 = \frac{\sin(3t)}{1}$$

$$i^2 9Ae^{i3t} + i3Ae^{i3t} + 2Ae^{i3t} = -ie^{i3t}$$

$$i^2 9A + i3A + 2A = -i$$

$$A(i^2 9 + i3 + 2) = -i$$

$$A = \frac{-i}{(i^2 9 + i3 + 2)} = -\frac{-i}{-7 + 3i} = \frac{-i(-7 - 3i)}{49 + 9} = \frac{-3}{58} + \frac{7i}{58}$$

$$x_p(t) = \left(\frac{-3}{58} + \frac{7i}{58}\right)\left(\cos(3t) + isin(3t)\right)$$

$$x_p(t) = \frac{-3}{58}\cos(3t) - \frac{7}{58}\sin(3t)$$

$$\lambda^2 + \lambda + 2 = 0$$

$$\lambda = \frac{-1 \pm \sqrt{1 - 8}}{2} = -\frac{1}{2} \pm \frac{i\sqrt{7}}{2}$$

Underdamped

$$x(t) = x_h + x_p = Ae^{\left(-\frac{1}{2}+\frac{i\sqrt{7}}{2}\right)t} + Be^{\left(-\frac{1}{2}-\frac{i\sqrt{7}}{2}\right)t} - \frac{3}{58}\cos(3t) - \frac{7}{58}\sin(3t)$$

Transient behavior is governed by the exponentials of the function, and the sinusoidal part governs the steady state behavior. This is because the exponential section tends towards zero as t approaches infinity.

b. When $F_{ext} = \sin\left(\frac{10t}{3}\right), m_3 = 1, b_2 = 0, k_1 = 3\ and\ k_4 = 6.$

$$\ddot{x_3} = -\frac{b_2}{m_3}\dot{x}_3 - \frac{k_4 + k_1}{m_3}x_3 + \frac{f_{ext}}{m_3}$$

$$\ddot{x_3} = -0 - \frac{9}{1}x_3 + \frac{\sin\left(\frac{10t}{3}\right)}{1}$$

$$\ddot{x} + 9x = \sin\left(\frac{10t}{3}\right)$$

$$i^2\frac{100}{9}Ae^{i\frac{10}{3}t} + 9Ae^{i\frac{10}{3}t} = -ie^{i\frac{10}{3}t}$$

$$A\left(9 - \frac{100}{9}\right) = -i$$

$$A = -\frac{-i}{-\frac{19}{9}} = \frac{9i}{19}$$

$$x_p(t) = \frac{9i}{19}\left(\cos\left(\frac{10t}{3}\right) + isin\left(\frac{10t}{3}\right)\right)$$

$$x_p(t) = -\frac{9}{19}\sin\left(\frac{10t}{3}\right)$$

$$\lambda^2 + 9 = 0$$

$$\lambda = \pm 3i$$

Underdamped

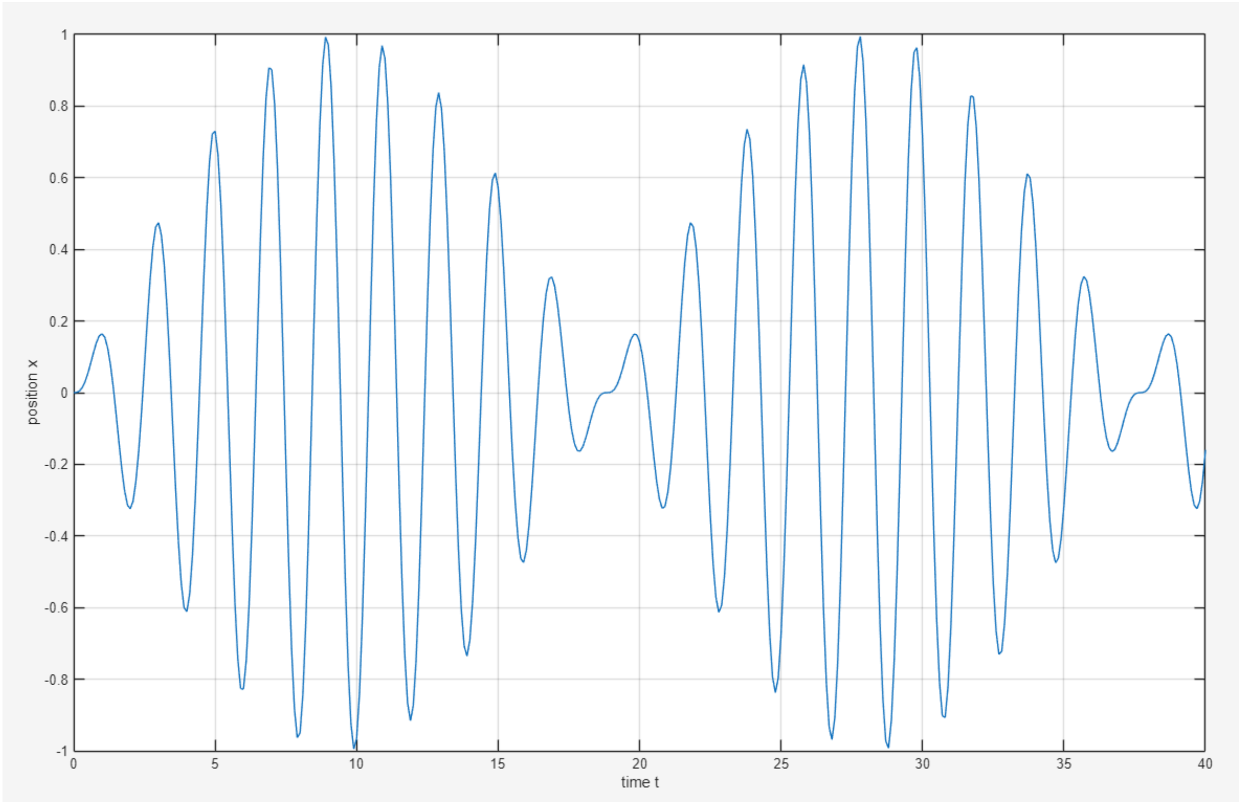
$$x(t) = x_h + x_p = Ae^{(3i)t} + Be^{(-3i)t} - \frac{9}{19}\sin\left(\frac{10t}{3}\right)$$

$$x(0) = A + B = 0$$

$$\dot{x} = 3iAe^{3it} - 3iBe^{-3it} - \frac{90}{57}\cos\left(\frac{10t}{3}\right) = 0$$

$$6iA = \frac{90}{57}; A = \frac{15}{57i}; B = -\frac{15}{57i}$$

$$x(t) = x_h + x_p = \frac{15}{57i}e^{(3i)t} - \frac{15}{57i}e^{(-3i)t} - \frac{9}{19}\sin\left(\frac{10t}{3}\right)$$



position x
time t
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
5
10
15
20
25
30
35
40

c. When $F_{ext} = \sin(3t)\,, m_3 = 1, b_2 = 0, k_1 = 6\ and\ k_4 = 3$

$$\ddot{x_3} = -\frac{b_2}{m_3}\dot{x}_3 - \frac{k_4 + k_1}{m_3}x_3 + \frac{f_{ext}}{m_3}$$

$$\ddot{x_3} = -0 - \frac{9}{1}x_3 + \frac{\sin(3t)}{1}$$

$$\ddot{x} + 9x = \sin(3t)$$

$$i^2 9Ae^{i3t} + 9Ae^{i3t} = -ie^{i3t}$$

$$A(i^2 9 + 9) = -i; e^{i3t}\ dosent\ work$$

$$\ddot{x} + 9x = \sin(3t)\,;$$

$$x = Ate^{i3t}$$

$$\dot{x} = Ae^{i3t} + i3Ate^{i3t}$$

$$\ddot{x} = i3Ae^{i3t} + i3A\left(e^{i3t} + i3te^{i3t}\right) = i6Ae^{i3t} - 9Ate^{i3t}$$

$$i6Ae^{i3t} - 9Ate^{i3t} + 9\left(Ate^{i3t}\right) = -ie^{i3t}$$

$$A(i6) = -i$$

$$A = \frac{-i}{i6} = -\frac{1}{6}$$

$$x_p = -\frac{1}{6}te^{i3t} = -\frac{1}{6}tcos(3t)$$

X homogenous see above

$$x(t) = x_h + x_p = Ae^{(3i)t} + Be^{(-3i)t} - \frac{1}{6}tcos(3t)$$

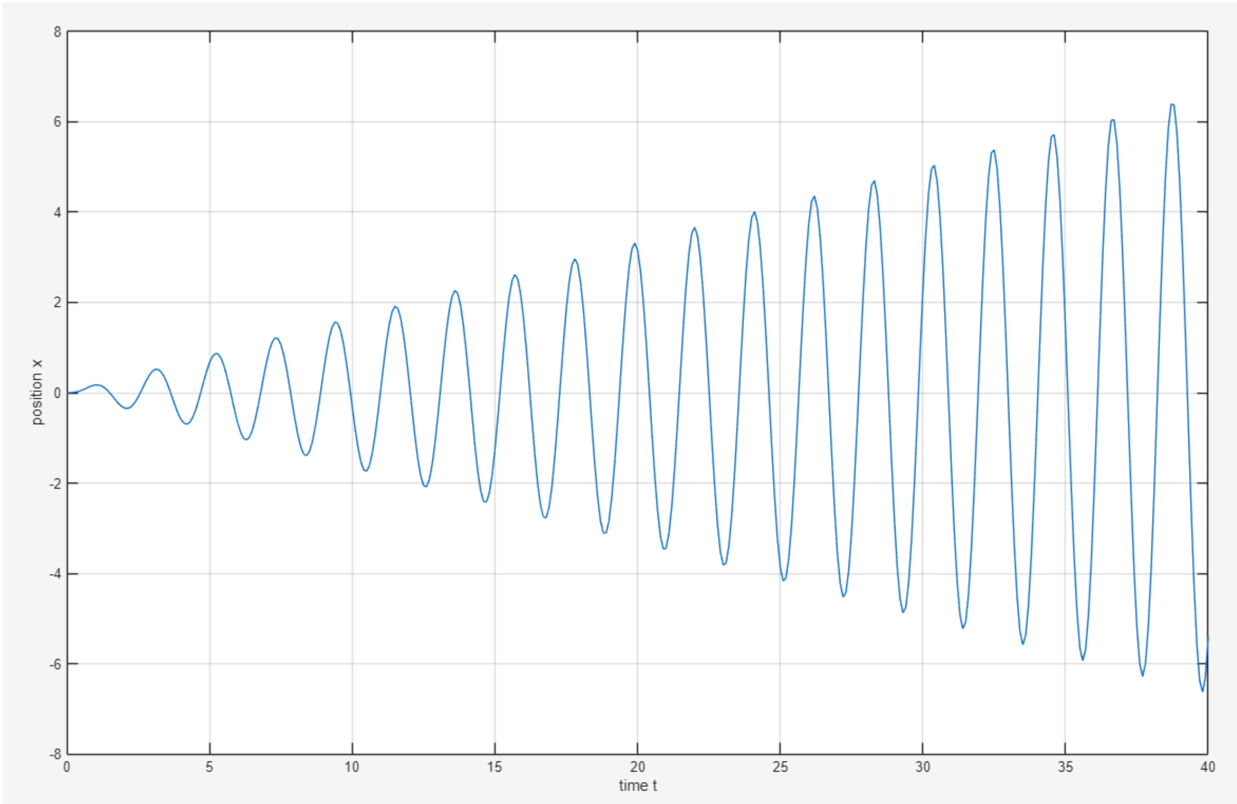
$$x(0) = A + B = 0$$

$$\dot{x} = A3ie^{3it} - Be^{-3it} + \frac{1}{2}tsin(3t) - \frac{1}{6}\cos(3t)$$

$$\dot{x}(0) = 3Ai - 3Bi - \frac{1}{6} = 0$$

$$6Ai = \frac{1}{6}; A = \frac{1}{36i}; B = -\frac{1}{36i}$$

$$x(t) = x_h + x_p = \frac{1}{36i}e^{(3i)t} - \frac{1}{36i}e^{(-3i)t} - \frac{1}{6}tcos(3t)$$


8
6
4
2
0
-2
-4
-6
-8
0
5
10
15
20
25
30
35
40
time t
position x

```
% Kyle Kasmiersky
% professor Luo - Vibes
% Homework 4
%-------------------------------------------------------------------------
clc; clear; close all

t = 0:0.1:40;

x1 = (15/(57*i)).*exp(3.*i.*t) - (15/(57*i)).*exp(-3.*i.*t) - (9/19).*sin(10.*t./3);

figure("Name","3b")

plot(t,x1)
grid on
xlabel("time t")
ylabel("position x")


% fig 2

x2 = exp(3.*i.*t)./(36*i) - exp(-3.*i.*t)./(36*i) - t.*cos(3.*t)./6;

figure('Name',"3c")

plot(t,x2)
grid on
xlabel("time t")
ylabel("position x")
```